

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456693.76235 +/- 0.00063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.014
Transit depth (Rp/Rs)^2	2.27 +/- 0.42 [%]
Orbital Inclination (inc)	81.51 +/- 0.91
Airmass coefficient 1 (a1)	1.151 +/- 0.011
Airmass coefficient 2 (a2)	-0.0878 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	None
Optimal Aperture	4.0
Optimal Annulus	15.52
Transit Duration (day)	0.0482 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.151 ± 0.014 vs 0.1594 (deviation 0.59σ)
Duration fit vs published	0.0482 d vs 0.0512 d (deviation 0.83σ)
Mid-time O-C	-1.8 min
Depth SNR	10.6
Residual scatter	0.94 %

Light curve

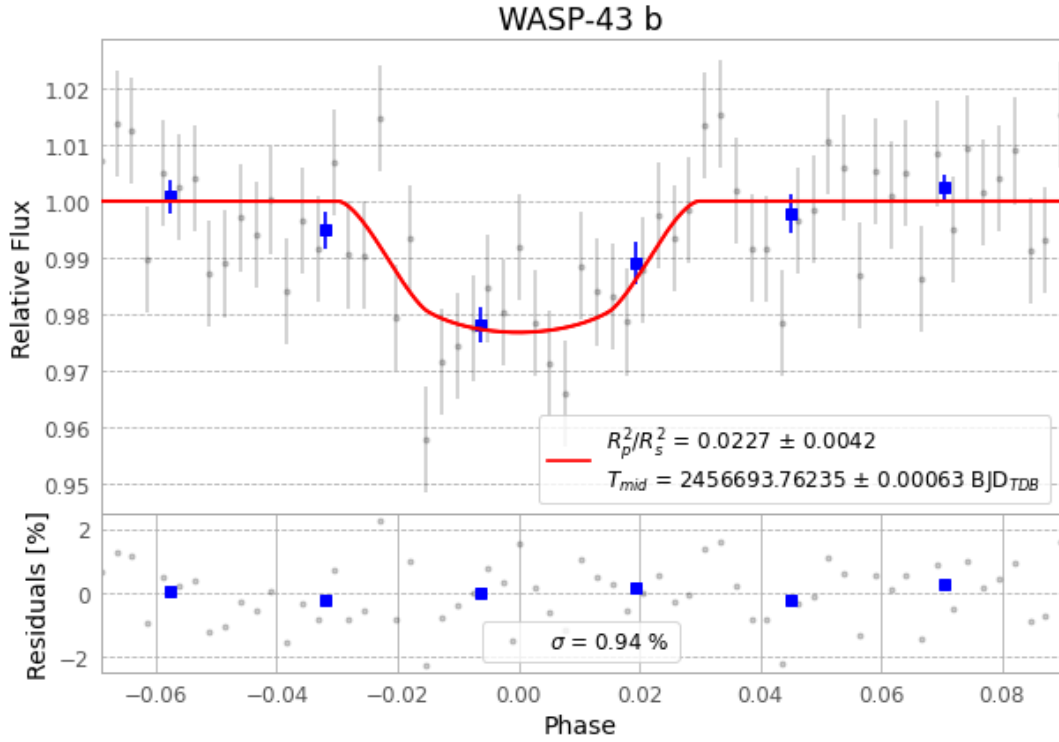


Figure 1 — FinalLightCurve_WASP-43 b_2014-02-04.png (SHA-256 3a0ed539792de618...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [118, 190], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c5ef065b73ef9f31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), WASP-43140205044812.FITS → WASP-43140205075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140205.log (ff7b24645634...), FinalLightCurve_WASP-43 b_2014-02-04.png (3a0ed539792d...), FinalLightCurve_WASP-43 b_2014-02-04.csv (d6fdb270b6d6...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:24Z.